

FLUARIX TETRA

Quadrivalent influenza vaccine (split virion, inactivated), suspension for injection

QUALITATIVE AND QUANTITATIVE COMPOSITION

Fluarix Tetra is an inactivated influenza vaccine (split virion), containing antigens (propagated in embryonated eggs) equivalent to the following strains:

A/Victoria/4897/2022 (H1N1)pdm09-like strain (A/Victoria/4897/2022, IVR-238);

A/Thailand/8/2022 (H3N2)-like strain (A/Thailand/8/2022, IVR-237);

B/Austria/1359417/2021-like strain (B/Austria/1359417/2021, BVR-26);

B/Phuket/3073/2013-like strain (B/Phuket/3073/2013, wild type).

This vaccine complies with the WHO recommended strains (Northern Hemisphere) for the season 2024/2025.

Each 0.5 mL vaccine dose contains 15 µg haemagglutinin of each of the recommended strains.

Fluarix Tetra meets the WHO requirements for biological substances and influenza vaccines and the European Pharmacopoeia requirements for influenza vaccines.

CLINICAL INFORMATION

Indications

Fluarix Tetra is a quadrivalent vaccine indicated for active immunisation of adults and children from 6 months of age for the prevention of influenza disease caused by influenza virus types A and B contained in the vaccine (see *Pharmacodynamics*).

The use of Fluarix Tetra should be based on official recommendations.

Dosage and Administration

Fluarix Tetra should be administered as a single 0.5 mL injection.

Children 6 months to less than 9 years of age who have not previously been vaccinated against influenza should receive a second dose of 0.5 mL after an interval of at least 4 weeks.

Children aged <6 months

The safety and efficacy of Fluarix Tetra in children aged <6 months have not been established.

Vaccination should be carried out by intramuscular injection preferably into the deltoid muscle or anterolateral thigh (depending on the muscle mass).

Contraindications

Fluarix Tetra should not be administered to subjects with known hypersensitivity after previous administration of Fluarix Tetra or influenza vaccines or to any component of the vaccine.

Warnings and Precautions

It is good clinical practice to precede vaccination by a review of the medical history (especially with regard to previous vaccination and possible occurrence of undesirable events) and a clinical examination.

As with all injectable vaccines, appropriate medical treatment and supervision should always be readily available in case of an anaphylactic event following the administration of the vaccine.

As with other vaccines, vaccination with Fluarix Tetra should be postponed in subjects suffering from an acute severe febrile illness. The presence of a minor infection, such as a cold, should not result in the deferral of vaccination.

It may be expected that in patients receiving immunosuppressive treatment or patients with immunodeficiency, an adequate immune response may not be elicited.

Fluarix Tetra is not effective against all possible strains of influenza virus. Fluarix Tetra is intended to provide protection against those strains of virus from which the vaccine is prepared and to closely related strains.

As with any vaccine, a protective immune response may not be elicited in all vaccinees.

FLUARIX TETRA SHOULD UNDER NO CIRCUMSTANCES BE ADMINISTERED INTRAVASCULARLY.

As with other vaccines administered intramuscularly, Fluarix Tetra should be given with caution to individuals with thrombocytopenia or any coagulation disorder since bleeding may occur following an intramuscular administration to these subjects.

Syncope (fainting) can occur following, or even before, any vaccination as a psychogenic response to the needle injection. It is important that procedures are in place to avoid injury from faints.

Interactions

Fluarix Tetra can be concomitantly administered with pneumococcal vaccines (see *Pharmacodynamics*).

If Fluarix Tetra is to be given at the same time as another injectable vaccine, the vaccines should always be administered at different injection sites.

False positive ELISA serologic tests for HIV-1, Hepatitis C, and especially HTLV-1 may occur following influenza vaccination. These transient false-positive results may be due to cross-reactive IgM elicited by the vaccine. For this reason, a definitive diagnosis of HIV-1, Hepatitis C, or HTLV-1 infection requires a positive result from a virus-specific confirmatory test (e.g. Western Blot or immunoblot).

Pregnancy and Lactation

The vaccine may be administered to pregnant women following an assessment of the risks and benefits.

The safety of Fluarix Tetra when administered to pregnant women has not been evaluated in clinical trials. When administered during pregnancy, safety data on inactivated seasonal influenza vaccines based on systematic literature review, and available post-marketing data on Fluarix Tetra, do not indicate an increased risk of adverse pregnancy outcomes.

Animal studies with Fluarix Tetra do not indicate direct or indirect harmful effects with respect to reproductive and developmental toxicity (see *Non-clinical Information*).

The safety of Fluarix Tetra when administered to breast-feeding women has not been evaluated. It is unknown whether Fluarix Tetra is excreted in human breast milk.

Fluarix Tetra should only be used during breast-feeding when the possible advantages outweigh the potential risks.

Adverse Reactions

Adverse reactions reported for Fluarix Tetra are listed according to the following frequency categories: Very common ($\geq 1/10$); Common ($\geq 1/100$ to $< 1/10$); Uncommon ($\geq 1/1,000$ to $< 1/100$); Rare ($\geq 1/10,000$ to $< 1/1,000$); Very rare ($< 1/10,000$).

Clinical trial data

Adults

A study with Fluarix Tetra in adults has evaluated the incidence of adverse reactions in subjects ≥ 18 years who received one dose of Fluarix Tetra (N = 3,036) or Fluarix (N = 1,010).

The following adverse reactions per dose have been reported:

Adverse Reactions	Frequency
Myalgia, injection site pain, fatigue	Very common
Headache, gastrointestinal symptoms (including nausea, vomiting, diarrhoea and/or abdominal pain), sweating ¹ , arthralgia, injection site redness, injection site swelling, shivering, fever, injection site induration ¹	Common
Dizziness ² , injection site hematoma ² , injection site pruritus ²	Uncommon

¹Reported in previous Fluarix trials

²Reported as unsolicited adverse reaction

Children aged 6 months to <18 years

Two clinical studies evaluated the reactogenicity and safety of Fluarix Tetra in children who received at least one dose of Fluarix Tetra or a control vaccine.

One study enrolled children 3 to <18 years of age who received Fluarix Tetra (N = 915) or Fluarix (N = 912). The second study enrolled children 6 to <36 months of age who received

Fluarix Tetra (N = 6,006) or a non-influenza vaccine control (N = 6,012) (see *Pharmacodynamics*).

The following adverse reactions per dose have been reported:

Adverse reactions	Frequency		
	6 to <36 months	3 to <6 years	6 to <18 years
Loss of appetite	Very common	Common	N/A
Irritability/Fussiness	Very common	Very common	N/A
Drowsiness	Very common	Common	N/A
Headache	N/A	N/A	Common
Gastrointestinal symptoms (including nausea, diarrhoea, vomiting and/or abdominal pain)	N/A	N/A	Common
Rash ¹	N/R	Uncommon	Uncommon
Myalgia	N/A	N/A	Very common
Arthralgia	N/A	N/A	Common
Fever ($\geq 38.0^{\circ}\text{C}$)	Common	Common	Common
Fatigue	N/A	N/A	Very common
Injection site pain	Very common	Very common	Very common
Injection site redness	Very common	Very common	Very common
Injection site swelling	Common	Very common	Very common
Shivering	N/A	N/A	Common
Injection site pruritus ¹	N/R	Uncommon	Uncommon
Injection site induration ²	N/A	Common	Common

N/A=Not solicited in this age group

N/R=Not reported

¹Reported as unsolicited adverse reaction

²Reported in previous Fluarix trials

Post-marketing data

The following adverse reactions have been observed for Fluarix and/or Fluarix Tetra during post-marketing surveillance.¹

Adverse reactions	Frequency
Transient lymphadenopathy, allergic reactions (including anaphylactic reactions), neuritis, acute disseminated encephalomyelitis, Guillain-Barré syndrome (GBS) ² , urticaria, pruritus, erythema, angioedema, influenza-like illness, malaise	Rare

¹Three of the influenza strains contained in Fluarix are included in Fluarix Tetra.

²Spontaneous reports of GBS have been received following vaccination with Fluarix and Fluarix Tetra; however, a causal association between vaccination and GBS has not been established.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamics

Mechanism of Action

Fluarix Tetra induces haemagglutination-inhibition (HI) antibodies against the 4 influenza virus strains contained in the vaccine. While specific levels of HI antibody in response to inactivated influenza virus vaccines have not been correlated with protection from influenza illness, the HI antibody titres have been used as a measure of vaccine activity. In some human challenge studies, HI antibody titres of $\geq 1:40$ have been associated with protection from influenza illness in up to 50% of subjects.

Annual revaccination with the current vaccine is recommended because immunity declines during the year after vaccination, and because circulating strains of influenza virus might change from year to year.

Pharmacodynamic Effects

Efficacy in children 6-35 months of age:

The vaccine efficacy (VE) of Fluarix Tetra was evaluated in study D-QIV-004, a randomised, observer-blind, non-influenza vaccine-controlled trial conducted during influenza seasons 2011 to 2014. Healthy subjects aged 6 through 35 months were randomised (1:1) to receive Fluarix Tetra (N = 6,006) or an age-appropriate non-influenza control vaccine (N = 6,012). Subjects were administered 1 dose (in case of history of influenza vaccination) or 2 doses, approximately 28 days apart.

VE of Fluarix Tetra was assessed for the prevention of influenza A and/or B disease (moderate to severe and of any severity) due to any seasonal influenza strain, starting 2 weeks post-vaccination until the end of the influenza season (approximately 6 months later). Fluarix Tetra met the predefined criteria for primary and secondary VE objectives (Table 1).

Table 1: Attack rates and VE in children 6-35 months of age (ATP (according to protocol) cohort for efficacy – time to event)

	Fluarix Tetra (N = 5,707)	Active comparator (N = 5,697)	Vaccine efficacy	
	Attack rate (%)	Attack rate (%)	%	CI
Any severity Influenza¹				
RT-PCR ³ confirmed	6.03	11.62	49.8	41.8; 56.8 ⁴
Culture confirmed	5.31	10.57	51.2	44.1; 57.6 ⁵
Culture confirmed vaccine matching strains	1.54	3.79	60.1	49.1; 69.0 ⁵
Moderate to Severe Influenza²				
RT-PCR ³ confirmed	1.58	4.25	63.2	51.8; 72.3 ⁴
Culture confirmed	1.38	3.79	63.8	53.4; 72.2 ⁵
Culture confirmed vaccine matching strains	0.35	1.54	77.6	64.3; 86.6 ⁵
Lower respiratory Illness RT-PCR Confirmed	0.49	1.07	54.0	28.9; 71.0 ⁵
Acute Otitis media RT PCR-confirmed	0.21	0.49	56.6	16.7; 78.8 ⁵

¹ Defined as an episode of influenza-like illness (ILI, i.e. fever $\geq 38^{\circ}\text{C}$ with any of the following: cough, runny nose, nasal congestion, or breathing difficulty) or a consequence of influenza virus infection [acute otitis media (AOM) or lower respiratory illness (LRI)].

² Defined as a subset of any influenza disease, with any of the following: fever $> 39^{\circ}\text{C}$, physician-diagnosed AOM, physician-diagnosed lower respiratory tract infection, physician-diagnosed serious extra-pulmonary complications, hospitalisation in the intensive care unit, or supplemental oxygen required for more than 8 hours.

³ reverse transcription polymerase chain reaction

⁴ 2-sided 97.5% confidence interval

⁵ 2-sided 95% confidence interval

Exploratory analyses were conducted on the Total Vaccinated Cohort (TVC) including 12,018 subjects. Fluarix Tetra was efficacious in the prevention of moderate to severe influenza caused by each of the 4 strains (Table 2), even when there was significant antigenic mismatch with 2 of the vaccine strains (A/H3N2 and B/Victoria).

Table 2: Attack rates and VE for RT-PCR confirmed moderate to severe disease by Influenza A subtypes and Influenza B lineages in children 6-35 months of age (TVC)

Strain	Fluarix Tetra (N = 6,006)	Active comparator (N = 6,012)	Vaccine Efficacy	
	Attack rate (%)	Attack rate (%)	%	95% CI
A/H1N1 ¹	0.22	0.77	72.1	49.9; 85.5
A/H3N2 ²	0.88	1.86	52.7	34.8; 66.1
B/Victoria ³	0.05	0.25	80.1	39.7; 95.4
B/Yamagata ⁴	0.37	1.21	70.1	52.7; 81.9

¹ to ⁴ Proportion of antigenic matching strains was 84.8%, 2.6%, 14.3% and 66.6%, for A/H1N1, A/H3N2, B/Victoria, and B/Yamagata, respectively.

Additionally, for RT-PCR confirmed cases of any severity, Fluarix Tetra reduced the risk of visits to the general practitioner by 47% (Relative Risk (RR): 0.53 [95% CI: 0.46; 0.61], i.e., 310 versus 583 visits) and to the emergency room by 79% (RR: 0.21 [95% CI: 0.09; 0.47], i.e., 7 versus 33 visits). The use of antibiotics was reduced by 50% (RR: 0.50 [95% CI: 0.42; 0.60],

i.e., 172 versus 341 subjects).

Immunogenicity in children and adults:

Immunogenicity of Fluarix Tetra was evaluated in terms of HI Geometric mean antibody titre (GMT) at 28 days after the last dose (children) or Day 21 (adults) and HI seroconversion rate (4-fold rise in reciprocal titre or change from undetectable [< 10] to a reciprocal titre of ≥ 40).

In study D-QIV-004, the evaluation was performed in a sub-cohort of 1,332 children (Table 3).

The effect of a 2-dose priming schedule in D-QIV-004 was evaluated by assessing the immune response after revaccination one year later with 1 dose of Fluarix Tetra in study D-QIV-009. This study demonstrated that 7 days post-vaccination, immune memory in children 6 to 35 months of age had been elicited for all 4 vaccine strains.

Immunogenic non-inferiority of Fluarix Tetra was assessed versus Fluarix in children (study D-QIV-003) and in adults (study D-QIV-008). Children received 1 or 2 doses and adults received 1 dose of either vaccine. In both studies, Fluarix Tetra elicited an immune response against the 3 strains in common that was non-inferior to Fluarix and a superior immune response against the additional B strain included in Fluarix Tetra (Table 3).

Table 3: Post-vaccination GMT and seroconversion rates (SCR) in children (6-35 months; 3 to < 18 years) and adults ≥ 18 years (ATP (95% CI))

Children 6 to 35 months of age (D-QIV-004)				
	Fluarix Tetra		Control	
	N=750-753	N'=742-746	N=578-579	N'=566-568
	GMT¹	SCR¹	GMT¹	SCR¹
A/H1N1	165.3 (148.6;183.8)	80.2% (77.2;83.0)	12.6 (11.1;14.3)	3.5% (2.2;5.4)
A/H3N2	132.1 (119.1;146.5)	68.8% (65.3;72.1)	14.7 (12.9;16.7)	4.2% (2.7;6.2)
B (Victoria)	92.6 (82.3;104.1)	69.3% (65.8;72.6)	9.2 (8.4;10.1)	0.9% (0.3;2.0)
B (Yamagata)	121.4 (110.1;133.8)	81.2% (78.2;84.0)	7.6 (7.0;8.3)	2.3% (1.2;3.9)
Children 3 to < 18 years (D-QIV-003)				
	Fluarix Tetra		Fluarix²	
	N=791	N'=790	N=818	N'=818
	GMT	SCR	GMT	SCR
A/H1N1	386.2 (357.3;417.4)	91.4% (89.2;93.3)	433.2 (401.0;468.0)	89.9% (87.6;91.8)
A/H3N2	228.8 (215.0;243.4)	72.3% (69.0;75.4)	227.3 (213.3;242.3)	70.7% (67.4;73.8)
B (Victoria)	244.2 (227.5;262.1)	70.0% (66.7;73.2)	245.6 (229.2;263.2)	68.5% (65.2;71.6)
B (Yamagata)	569.6 (533.6;608.1)	72.5% (69.3;75.6)	224.7 (207.9;242.9)	37.0% (33.7;40.5)
Adults ≥ 18 years (D-QIV-008)				
	Fluarix Tetra		Fluarix²	
	N=1,809	N'=1,801	N=608	N'=605
	GMT	SCR	GMT	SCR
A/H1N1	201.1 (188.1;215.1)	77.5% (75.5;79.4)	218.4 (194.2;245.6)	77.2% (73.6;80.5)
A/H3N2	314.7 (296.8;333.6)	71.5% (69.3;73.5)	298.2 (268.4;331.3)	65.8% (61.9;69.6)

B (Victoria)	404.6 (386.6;423.4)	58.1% (55.8;60.4)	393.8 (362.7;427.6)	55.4% (51.3;59.4)
B (Yamagata)	601.8 (573.3;631.6)	61.7% (59.5;64.0)	386.6 (351.5;425.3)	45.6% (41.6;49.7)

N = Number of subjects with post-vaccination results available (for GMT)

N' = Number of subjects with both pre- and post-vaccination results available (for SCR)

¹results from the immunogenicity sub-cohort

²B (Yamagata) strain was not included in Fluarix

Concomitant administration with pneumococcal vaccines:

In clinical study D-QIV-010 involving 356 adults ≥ 50 years of age at risk for complications of influenza and pneumococcal diseases, subjects received Fluarix Tetra and 23-valent pneumococcal polysaccharide vaccine (PPV23) either concomitantly or separately. For all 4 Fluarix Tetra vaccine strains and the 6 pneumococcal serotypes (1, 3, 4, 7F, 14, and 19A) in PPV23 evaluated in the pre-specified primary analysis, the immune response was non-inferior between the 2 groups. Based on a descriptive analysis for 6 additional pneumococcal vaccine serotypes (5, 6B, 9V, 18C, 19F, and 23F), the immune response was comparable between groups, with 91.7% to 100% and 90.7% to 100% of subjects attaining seroprotective antibody levels against these serotypes in the separate and concomitant administration group respectively.

Immunological non-inferiority has been demonstrated based on published data for all 3 Fluarix trivalent strains (D-TIV) and all 13-valent pneumococcal conjugate vaccine (PCV13) serotypes in adults 50-59 years of age, as well as for 2 of 3 D-TIV strains and 12 of 13 PCV13 serotypes in adults >65 years of age. A lower immune response to some pneumococcal serotypes was observed when PCV13 was given concomitantly with D-TIV as compared to separate administration, however the clinical relevance of this observation is unknown.

Non-Clinical Information

Non-clinical data reveal no special hazards for humans based on conventional studies of acute toxicity, local tolerance, repeated dose toxicity and reproductive/developmental toxicity.

PHARMACEUTICAL INFORMATION

List of Excipients

Sodium chloride, disodium phosphate dodecahydrate, potassium dihydrogen phosphate, potassium chloride, magnesium chloride hexahydrate, α -tocopheryl hydrogen succinate, polysorbate 80, octoxinol 10 and water for injections.

Hydrocortisone, gentamicin sulphate, ovalbumin, formaldehyde and sodium deoxycholate are present as residues from the manufacturing process.

Shelf Life

The expiry date is indicated on the label and packaging.

Storage

Store at 2°C - 8°C (in a refrigerator).

Do not freeze.

Store in the original package in order to protect from light. The storage conditions are detailed on the packaging.

Nature and Contents of Container

0.5 ml of suspension in a pre-filled syringe (type I glass) with a plunger stopper (butyl rubber) and with a rubber tip cap.

The tip cap and rubber plunger stopper of the pre-filled syringe are not made with natural rubber latex.

Pack sizes of 1 and 10, with or without needles.

Not all presentations are available in every country.

Incompatibilities

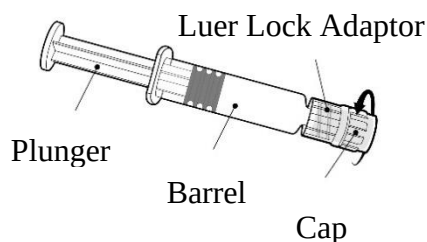
In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

Use and Handling

The vaccine presents as a colourless to slightly opalescent suspension.

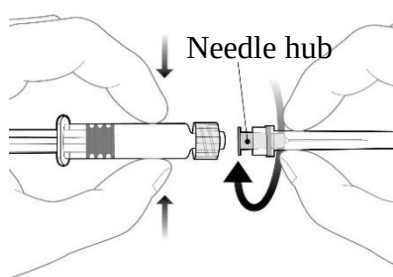
The syringe should be shaken and inspected visually for any foreign particulate matter and/or variation of physical aspect prior to administration. In the event of either being observed, discard the vaccine.

Instructions for the pre-filled syringe



Hold the syringe by the barrel, not by the plunger.

Unscrew the syringe cap by twisting it anticlockwise.



To attach the needle, connect the hub to the Luer Lock Adaptor and rotate a quarter turn clockwise until you feel it lock.

Do not pull the syringe plunger out of the barrel. If it happens, do not administer the vaccine.

Disposal:

Any unused product or waste material should be disposed of in accordance with local requirements.

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Manufacturer responsible for batch release:

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